Sashank Solasa

Software Development Engineer

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Education

B.Tech in Computer Science and Engineering 2023 – 2027
VIT-AP University, Amaravati *CGPA: 8.53 / 10.0*

Class XII (BIEAP) 2023
Sri Chaitanya Junior College, Vijayawada *96.5%*

Class X (BSEAP) 2021
Sri Chaitanya Techno School, Devarapalli *99%*

Experience

Full-Stack Developer (Freelance), VPSM Jun 2026 – Jul 2026

- Built a supermarket POS and inventory system with barcode checkout, billing, stock tracking, supplier management, purchase orders and sales reports.
- Wrote a Spring Boot service that forecasts product demand, flags low-stock and slow-moving items, and suggests restock quantities.
- Added an LLM assistant so staff can ask sales and stock questions in plain English instead of digging through reports.

Projects

DijkMap | Spring Boot, PostgreSQL, PostGIS, Docker Jun 2026 – Oct 2026 github.com/solasas/DijkMap

- Built an OpenStreetMap-based route planner with a custom min-heap Dijkstra implementation over a 509K-node, 1.0M-edge road graph, computing 150 km inter-city routes in approximately 200 ms.
- Implemented bidirectional Dijkstra, reducing node expansions by 27–33% across benchmarks, and Yen's k-shortest paths for alternative routes, alongside an isochrone API endpoint.
- Integrated LLM-based natural-language routing preferences, translating requests into hard constraints and soft preferences while preserving hard constraints and flagging requirements unverifiable from available map data.

NetRace | Spring Boot, React, Vite Sep 2026 github.com/solasas/NetRace

- Built a website performance analyzer that takes a URL and breaks its load time into measured stages: DNS lookup, TCP connection, TLS handshake, time to first byte, and download.
- Ensured every figure comes from a real request to the site, with nothing guessed or estimated, and added SSRF protections so the tool cannot be used to reach internal or private networks.
- Built a waterfall chart for single-site breakdowns and a comparison view for up to 5 sites to show which is faster and why.

Paraspa | Flutter, Spring Boot, PostgreSQL, JWT, WebSocket Apr 2026 github.com/solasas/paraspa

- Peer-to-peer skill exchange app: users list skills they can teach and want to learn, get matched with compatible peers, and manage swap requests, sessions and ratings.
- Backend has 40+ REST endpoints covering matching, search, session scheduling and rescheduling, and ratings.
- Real-time 1-to-1 chat over WebSocket with saved history and unread counts. JWT auth and access checks keep users limited to their own exchanges and sessions.
- Runs in Docker Compose; service layer covered by 60+ JUnit and Mockito tests.

Technical Skills

- **Languages:** Java, Python, JavaScript
- **Backend & Frameworks:** Spring Boot, Spring Data JPA, REST APIs, JWT, WebSocket, Spring AI, React, Flutter
- **Databases:** PostgreSQL, MySQL, Redis, MongoDB
- **Cloud (AWS):** EC2, S3, RDS, Lambda, IAM, DynamoDB
- **Tools & Build:** Git, GitHub, Maven, JUnit, Mockito, Postman, Docker, Kubernetes, Kafka, CI/CD, JMeter
- **Data & AI/ML:** NumPy, Pandas, Scikit-learn, ETL, LLMs, RAG
- **Core CS:** Data Structures and Algorithms, OOP, DBMS, Operating Systems, Computer Networks, System Design

Certifications

- **AWS Cloud Foundations** — AWS Academy
- **Docker** — Udemy
- **PostgreSQL** — Udemy
- **Java Spring Framework, Spring Boot** — Udemy
- **Generative AI** — IBM

Positions of Responsibility

- **ACM Student Chapter, VIT-AP:** Technical team core member; helped run technical events and workshops.
- **Photography Club, VIT-AP:** Core member; ran photography workshops and photowalks.